the same relationship as between electromagnetic waves and photons. The waves give an approximate large-scale vision of photons. Photons are the way in which waves interact. Continuous space and time are an approximate large-scale vision of the dynamic of quanta of gravity. The quanta of gravity are the way in which space and time interact. The same mathematics coherently describes the quantum gravitational field as other quantum fields.

The conceptual price paid is the relinquishing of the idea of space, and of time, as general structures within which to frame the world. Space and time are approximations which emerge at a large scale. Kant was perhaps right when he affirmed that the subject of knowledge and its object are inseparable, but he was definitely mistaken when he considered Newtonian space and time as a priori forms of knowledge, parts of an indispensable grammar for understanding the world. This grammar has evolved, and is still in the process of evolving, with the growth of our knowledge.

General relativity and quantum mechanics are, in the end, not as incompatible as they seemed. On closer inspection, they shake hands and engage in a beautiful dialogue. The spatial relations that weave Einstein's curved space are the very interactions weaving the relations between the systems of quantum mechanics. The two become compatible and conjoined, two sides of the same coin, as soon as it is recognized that space and time are aspects of a quantum field, and quantum fields can exist even without being grounded in an external space.

This rarefied picture of the fundamental structure of the physical world is the vision of reality offered today by quantum gravity.

The main reward of this kind of physics is that, as we shall see in the next chapter, infinity disappears. The infinitely small no longer exists. The infinities which plague conventional quantum field theory, predicated on the notion of a continuous space, now vanish, because they were generated precisely by the assumption, physically incorrect, of the continuity of space. The singularities which render Einstein's equations absurd when the gravitational field becomes too strong also disappear: they are only the result of neglecting the quantization of the field. Little by little, the pieces of the puzzle find their place. In the final sections of this book, I describe some of the physical consequences of this theory.

It may appear strange and difficult to think of discrete elementary entities not in space and time, but weaving space and time with their relations. But how strange it must have seemed to listen to Anaximander, when he claimed that beneath our feet there was only the same sky that we can see above our heads? Or to Aristarchus, when he tried to measure the distance from the Earth to the Moon and the Sun, discovering that they are extremely distant, and are therefore not the size of little balls, but gigantic – and the Sun is immense compared to the Earth. Or to Hubble, when he realized that the